Probing Intuitive Physics in Pretrained Video Foundation Models

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Context

This project studies whether pretrained video foundation models encode intuitive physics, and compares three pretraining paradigms under the same frozen-feature probing setup: Stable Video Diffusion (SVD), V-JEPA, and VideoMAE / VideoMAE-v2. The central question is whether physical knowledge is more accessible in generative, predictive, or masked-reconstruction video representations. We focus on probing rather than finetuning, so the evaluation reflects what is already present in the pretrained features. This is well motivated by recent benchmarks such as IntPhys 2, which tests intuitive physics in videos, and MVP, which was designed to reduce shortcut-based inflation using minimal video pairs [1, 2].

Project

Datasets and Protocol The main benchmark will be IntPhys 2, which evaluates four core physical principles: Permanence, Immutability, Spatio-Temporal Continuity, and Solidity. It is specifically designed to distinguish physically possible from impossible events in controlled environments, making it a strong benchmark for probing physical understanding rather than generic action recognition [1].

As a second benchmark, we will use MVP (Minimal Video Pairs), which contains around 55K multiple-choice video-QA examples and pairs visually similar videos with the same question but opposite answers. This makes it much harder for a model to rely on superficial visual or language shortcuts [2].

To separate physics understanding from generic video recognition, we will add one standard motion-heavy video benchmark, such as Something-Something v2, as a control task. This will help determine whether gains on physics benchmarks simply reflect better temporal modeling.

Models We will compare one pretrained model from each family:

1. Stable Video Diffusion model, used as a frozen feature extractor and optionally also as a black-box plausibility scorer;
2. V-JEPA, a predictive self-supervised video model aimed at understanding, prediction, and planning in the physical world;
3. VideoMAE-v2 or VideoMAE, a strong masked video autoencoding baseline.

This comparison is meaningful because the three models represent distinct learning objectives: generative denoising, predictive representation learning, and masked reconstruction [3–5].

Feature Extraction and Layer Selection The main experiment will use layerwise probing. For all transformer backbones, we will extract features from four depths: approximately 25%, 50%, 75%, and 100% of the encoder. For a 24-block model this corresponds to blocks 6, 12, 18, 24; for a 12-block model it corresponds to 3, 6, 9, 12. This keeps the comparison fair across architectures.

For V-JEPA, this choice is especially natural because recent V-JEPA evaluation practice reports gains from combining the last layer with several intermediate layers, rather than relying only on the final representation [3, 4].

For VideoMAE / VideoMAE-v2, we will use the same four relative depths for fairness, even if many prior evaluations emphasize the final-layer representation. VideoMAE-v2 is a strong and scalable baseline for general video representation learning, so using the same protocol avoids giving one family an unfair advantage [5].

For the video diffusion model, we will probe both multiple layers and multiple noise levels, since information in diffusion models may depend strongly on denoising stage. A practical choice is to evaluate three diffusion regimes: early/high noise, middle, and late/low noise, such as roughly 90%, 50%, and 10% of the denoising schedule, combined with the same four backbone depths.

Probe Design We will compare three probe families of increasing strength:

1. a linear probe on pooled clip features,
2. a 2-layer MLP,
3. a temporal attentive probe operating over frame or token sequences.

This is important because a weak linear probe may underestimate what a representation contains. If a model performs poorly with a linear probe but well with an attentive probe, then the physics information is present but less explicitly organized. This is especially relevant for V-JEPA-style frozen evaluation, where more expressive probing has been shown to improve results on temporally demanding tasks [3, 4].

How the Datasets Will Be Used For IntPhys 2, the cleanest setup is binary classification or pairwise ranking: given a matched possible/impossible pair, the probe predicts which clip is physically valid. We will also report performance by principle: Permanence, Immutability, Continuity, and Solidity.

For MVP, the model should be evaluated on pair consistency, not only per-example accuracy. Since each example has a minimal paired counterpart with the opposite answer, this gives a better measure of whether the probe actually captures physical reasoning rather than shortcuts [2].

For the control video-classification dataset, we will keep the same clip length, temporal sampling, and aggregation strategy across all backbones.

Controls and Expected Outcomes We will include a single-frame baseline and a time-shuffled baseline. If a model retains high "physics" performance after temporal shuffling, then it is likely exploiting static cues rather than temporal physical reasoning. We expect all three model families to contain some physical information, but with different accessibility: VideoMAE may be strongest on generic video understanding, V-JEPA may expose more predictive/world-model structure, and video diffusion may reveal useful physical structure either through probing or through a direct plausibility score. The most interesting outcome is not only which model performs best, but which layers and which probe strengths reveal intuitive physics most clearly across the three pretraining paradigms.

References

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